

STADIOcare Request

Purpose: Data request for a model that predicts the likelihood of a 30-day readmission at the time of discharge. STADIOcare should provide historical patient, hospital encounter, clinical, and discharge data. The requested data should cover a sufficiently long period to capture different patient groups and admission patterns. Where possible, data should be provided at patient and admission level using a unique, anonymised patient identifier.

Coding requirement: For every coded or categorical field, STADIOcare should provide the complete list of valid values/codes and their meanings in a data dictionary or mapping table. The project will not assume or invent STADIOcare coding values.

1. Patient and Demographic Data

Column name	Data type	Example	Additional information
Patient ID	String	P122445	Unique anonymised identifier; consistent across encounters.
Age	Integer	71	Age at admission
Gender	Categorical	Female	Provide all valid values
Chronic conditions	Categorical / multiple values	Diabetes; Hypertension	Existing diagnosed chronic conditions available before discharge.
Province / geographic area	Categorical	Gauteng	If available and approved; provide valid values.
Medical aid / funding type	Categorical	Medical aid	If available and approved

2. Hospital Encounter and Admission Data

Column name	Data type	Example	Additional information
Admission ID	String	A122	Unique identifier for each hospital encounter.
Patient ID	String	P122445	Links encounter to anonymised patient.
Admission date/time	Date/Time	2026-03-14 09:25:00	Use YYYY-MM-DD HH:MM:SS.
Discharge date/time	Date/Time	2026-03-18 14:10:00	Use the same date/time format.
Admission type	Categorical	Emergency	Provide all valid values/codes and definitions.
Admission source	Categorical	Emergency department	Provide all valid values .
Discharge disposition	Categorical	Home	Patient destination/status at discharge; provide all valid values.
Length of stay	Numeric	4 days	Can be derived from admission and discharge date/time.
ICU admission flag	Binary	1 = Yes	Whether patient entered ICU during encounter, if available.

ICU days	Numeric	2	Number of ICU days, if available.
----------	---------	---	-----------------------------------

3. Clinical, Diagnosis, Procedure and Treatment Data

Column name	Data type	Example	Additional information
Primary diagnosis code	Categorical	DX1023	Provide code plus coding system/version.
Primary diagnosis name	Categorical	Type 2 Diabetes	Mapped diagnosis description/name.
Secondary diagnosis codes	Categorical / multiple values	DX2041; DX3098	All additional diagnoses provide codes and names.
Secondary diagnosis names	Categorical / multiple values	Hypertension; Asthma	Mapped descriptions for secondary diagnosis codes.
Number of diagnoses	Integer	5	Total recorded diagnoses for encounter.
Procedure codes	Categorical / multiple values	PROC1023	All procedures performed before prediction point.
Procedure names	Categorical / multiple values	CT scan	Procedure descriptions/names.
Number of procedures	Integer	3	Can be derived from procedure records.
Medication names	Categorical / multiple values	Metformin	Medication names.
Medication category	Categorical	Antidiabetic	Provide categories if available.
Number of medications	Integer	7	Total medications associated with encounter.
Laboratory test code/name	Categorical	HbA1c	Include test name/code and coding system.
Laboratory result	Numeric / categorical	7.2	Include value, unit, reference range and timestamp where available.
Clinical severity indicator	Numeric / categorical	Severity level 3	If available; provide definition and valid values.
Complication indicator	Binary	1 = Yes /0=No	Complication recorded before prediction point, if available.

4. Previous Healthcare Utilisation and Readmission History

Column name	Data type	Example	Additional information
Previous admissions	Integer	4	Number of previous hospital admissions in a defined historical period.
Previous readmissions	Integer	1	Number of previous readmissions in a defined historical period.
Previous outpatient visits	Integer	3	Number of outpatient/clinic visits in a defined historical period.
Previous emergency visits	Integer	2	Number of emergency visits in a defined historical period.
Previous ICU admissions	Integer	1	Number of previous ICU admissions, if available.
Days since previous admission	Numeric	45 days	Derived from encounter history.
Days since previous discharge	Numeric	42 days	Derived from encounter history.
Previous admission diagnoses	Categorical / multiple values	Heart failure	Historical diagnoses; provide codes and mapped names.
Previous readmission diagnoses	Categorical / multiple values	Pneumonia	Diagnosis associated with previous readmissions; provide codes and names.
Previous procedures	Categorical / multiple values	PROC2020	Historical procedures with codes and mapped names.
Previous length of stay	Numeric	6 days	Derived from historical admission/discharge dates.

5. Readmission Outcome Data

Column name	Data type	Example	Additional information
Readmission within 30 days	Binary / categorical	1 = Yes; 0 = No	Target variable: readmission within 30 days after discharge.
Readmission date/time	Date/Time	2026-04-02 11:30:00	Subsequent admission date/time, where applicable.
Readmission type	Categorical	Emergency	Provide all valid values/codes and definitions.
Readmission diagnosis name	Categorical	Heart failure	diagnosis description/name.
Planned readmission flag	Binary / categorical	1 = Yes	Identify planned/elective

			readmissions separately.
Days to readmission	Numeric	18 days	Derived from discharge and readmission date/time.

6. Coding, Mapping and Data Dictionary Requirements

Column name	Data type	Example	Additional information
Diagnosis code mapping	Reference table	DX1023 -Type 2 Diabetes	Code, diagnosis name, category.
Data dictionary	Document	Field definitions	Definition, data type, valid values, coding system, units, missing value rules and source for every field.

7. Data Quality, Timing and Structure Requirements

Column name	Data type	Example	Additional information
Prediction timestamp	Date/Time	2026-03-18 14:10:00	Define exact prediction point. Predictors must be available by this point.
Date/time format	Standard	YYYY-MM-DD HH:MM:SS	Use one consistent format and document timezone if relevant.
Patient-level linkage	String	P122445	Same anonymised patient ID across encounters for longitudinal history.
Encounter-level linkage	String	A122	Unique admission/encounter ID for each episode.
Data coverage period	Date range	2022-01-01 to 2026-12-31	Specify full historical period and any collection changes.